

Asymptotic Analysis of Proportional Fair Algorithm

Jack M. Holtzman
Qualcomm, Inc.
5775 Morehouse Drive, San Diego, Ca 92121-1714
jholtzma@qualcomm.com

Abstract

It was previously shown (under simplifying assumptions) that this algorithm gives equal power and time to users who only differ in the distance from the BS, their fading characteristics being the same. We give here a more general result for two classes of users with different fading characteristics. All other things being equal, the user class with more fading variability gets more throughput with a lower (but not much lower) fraction of time transmitting. We also demonstrate a separability property – the effects of variability and distance from the base station are separable.

I. Introduction

In [1] - [3], it is shown how knowledge of the channel can increase CDMA capacity by transmitting mostly when the channel is good. An issue that arises in cellular CDMA is that of fairness because users nearer the base station (BS) would be favored in such approaches. There is thus a tradeoff between total throughput and fairness among users.

An algorithm based on priority given just by the C/I would always give all the power to the user close to the BS with the best channel, maximizing system throughput but also being unfair to users far from the BS. An algorithm was recently introduced which strikes a good compromise between throughput and fairness [4]. This algorithm (called a proportional fair algorithm) includes throughput monitoring which introduces

fairness – a user which does not get transmitted for too long gets its priority raised.

In [5] we presented a result which yields a simple interpretation of this algorithm showing how it provides a throughput leveling and an appealing tradeoff between throughput and fairness. We provide here an asymptotic analysis of the algorithm for two classes of users, which includes the interpretation of [5] as a special case. All other things being equal, the user class with more fading variability gets more throughput with a lower (but not much lower) fraction of time transmitting. We also demonstrate a separability property – the effects of variability and distance from the base station are separable.

There are a number of other papers on downlink scheduling, e.g., [6] which has further references.

2. Proportional Fair Algorithm

A simplified version is as follows: Let

$P(t)$ = available power at time t for use by K delay-tolerant data users.

$T_k(t)$ = throughput of user k over a time window up to time t , $k=1, \dots, K$

The user with highest priority in a time slot defined by

$$\frac{(C/I)_k(t)}{T_k(t)} \quad (1)$$

transmits with power $P(t)$.

The throughput monitoring displayed in (1) introduces fairness -- a user which does not transmit for too long because of a relatively low C/I gets its priority raised by the low denominator.

Remark on C/I. It is implicitly assumed that the supportable data rate is proportional to C/I, which is usually a reasonable CDMA approximation for not too high C/I. It is not accurate when multiple modulations or codings are used. This will be generalized in Section 4.

3. Simple Interpretation of Algorithm from [5]

Let

$$(C/I)_k(t) = a_k \cdot b_k(t), \quad k=1, \dots, K$$

a_k = distance dependent component of C/I for user k

$b_k(t)$ = random component of C/I for user k, including Rayleigh or Ricean fading.

Assumption 1: the $b_k(t)$ are iids among the users and from time slot to time slot, and they are all independent of P(t)

Remark. It is assumed that there is one slot for each user's transmission. This may not always be the case – lower data rate users might use more slots.

Assumption 2: Rate is (approximately) linear with power (continuous with no limit).

T_k = stationary throughput of user k (with the throughput window $W \rightarrow \infty$; in actuality the window would be chosen to do some averaging and short enough to control delays).

It is shown in [5] that, under Assumptions 1 and 2, T_k/a_k is the same for all k, and all users get transmitted with the same asymptotic fraction of time and with the same average power.

4. Analysis for Two Classes of Users

Consider two classes of users defined by

$$(C/I)_{1,j} = a_1 b_{1,j}(t), \quad j = 1, \dots, N_1$$

$$(C/I)_{2,j} = a_2 b_{2,j}(t), \quad j = 1, \dots, N_2$$

All the $\{b_{1,j}(t)\}$ and $\{b_{2,j}(t)\}$ are iid within each class and independent across classes (also independent across time slots). This is in contrast to [5] where all users' random components were iid's.

Now consider the window of observation for throughput monitoring $\rightarrow \infty$ so that T_1 and T_2 are considered converged to their stationary values. We are interested in determining the ratio of throughputs,

$$r = \frac{T_1}{T_2}$$

A user from Class 1 is chosen for transmission if

$$\frac{X_1}{T_1} > \frac{X_2}{T_2}$$

where

$$X_1 = \max_{j=1, \dots, N_1} \{a_1 b_{1,j}(t)\}$$

$$X_2 = \max_{j=1, \dots, N_2} \{a_2 b_{2,j}(t)\}$$

Then, since the transmitted rate of a class 1 user is proportional to X_1 , the total mean throughput of class 1 users is proportional to

$$P\left\{\frac{X_1}{T_1} > \frac{X_2}{T_2}\right\} E\left[X_1 \mid \frac{X_1}{T_1} > \frac{X_2}{T_2}\right]$$

and a similar expression for class 2. This leads to the ratio of total class throughputs:

$$\frac{T_1 N_1}{T_2 N_2} = \frac{P\{X_1 > r X_2\} E[X_1 \mid X_1 > r X_2]}{P\{X_2 > r^{-1} X_1\} E[X_2 \mid X_2 > r^{-1} X_1]}$$

which is a fixed point equation

$$r = f(r)$$

The numerator and denominator in the fixed point equation are evaluated as:

$$N_2 \int_0^{\infty} p_2(x_2) dx_2 \int_{r x_2}^{\infty} x_1 p_1(x_1) dx_1$$

and

$$N_1 \int_0^{\infty} p_1(x_1) dx_1 \int_{x_2/r}^{\infty} p_2(x_2) dx_2$$

where the $p_i(x)$ are the pdf's. (The properties of the order statistics are given in [7]).

Remark on uniqueness of fixed point. Since $f(r)$ is positive and monotonically decreasing for positive r , there is a unique fixed point for positive r . This is consistent with the uniqueness of the optimum to the concave optimization problem in [4].

Remark on the special case $b_1(t)$ and $b_2(t)$ being identically distributed. It is easily shown that the solution to the fixed point equation is

$$\frac{T_1}{T_2} = \frac{a_1}{a_2}$$

for $N_1 = N_2$ as found in [5].

Numerical examples

The first class is Rayleigh distributed and the second class is Ricean. For the Ricean pdf, we use $p(x)$ as given in (2.44) of [8]:

$$p(x) = \frac{2x(K+1)}{\Omega_p} \exp \left\{ -k - \frac{(K+1)x^2}{\Omega_p} \right\} \cdot I_0 \left(2x \sqrt{\frac{K(K+1)}{\Omega_p}} \right)$$

where here K is the Ricean factor (linear, not in dB), the ratio of powers of specular and scattered components and Ω_p is the average power.

Results are shown in Figures 1 and 2 for

$$a_1 = a_2 = 1, N_1 = N_2 = 5, K: 0 \rightarrow 10 \text{ (linear, not dB)}$$

and in Figures 3 and 4 for

$$a_1 = 1, a_2 = 10, N_1 = N_2 = 5, K: 0 \rightarrow 10 \text{ (linear, not dB)}$$

Note that in Figures 3 and 4, the result of [5] is recovered at $K=0$ (see the Remark preceding the numerical example). That is, the throughputs are, in that case, in proportion to the average C/I 's.

4. Generalization

One direction of generalization is to no longer assume that the supportable data rate is proportional to the C/I , but rather a function of it

$$R = f((C/I)) = g(ab(t))$$

Then a user's priority is

$$R_{i,j}/T_i = g(a_i b_{i,j}(t))/T_i$$

which is, in fact, the actual proportional fairness algorithm. Then, the fixed point equation becomes

$$r = \frac{P\{g(X_1) > r g(X_2)\} E[f(X_1) | g(X_1) > r g(X_2)] N_2}{P\{g(X_2) > r g(X_1)\} E[g(X_2) | g(X_2) > r g(X_1)] N_1}$$

5. Separability of Distance and Variability Effects

It was shown above (under simplifying assumptions) that this algorithm

- Gives equal power and time to users who only differ in the distance from the BS, their fading characteristics being the same.
- Gives larger throughput to users whose variability is greater than others' variability, with their distances to the BS being the same. The users with greater variability use a lower (but not much lower) fraction of time transmitting.

Here, we show that these effects are separable. That is, the combined effect is the superposition of one effect upon the other, with each determined independently. Recapitulating the results,

- When $b_1(t)$ and $b_2(t)$ are identically distributed,

$$\frac{T_1}{T_2} = \frac{a_1}{a_2}$$

and the two user classes get the same fraction of time transmitting.

- When $a_1 = a_2$, the user class with more variability gets more throughput (and a little smaller fraction of time transmitting).

To show separability of these effects, the fixed point equation is

$$r = \frac{P\{X_1 > rX_2\} E[X_1 | X_1 > rX_2] N_2}{P\{X_2 > r^{-1}X_1\} E[X_2 | X_2 > r^{-1}X_1] N_1} \times \frac{P\left\{\max_{j=1, \dots, N_1} \{b_{1j}(t)\} > r \frac{a_2}{a_1} \max_{j=1, \dots, N_1} \{b_{2j}(t)\}\right\} N_2 a_1}{P\left\{\max_{j=1, \dots, N_2} \{b_{2j}(t)\} > r^{-1} \frac{a_1}{a_2} \max_{j=1, \dots, N_1} \{b_{1j}(t)\}\right\} N_1 a_2} \times \frac{E\left[\max_{j=1, \dots, N_1} \{b_{1j}(t)\} \mid \max_{j=1, \dots, N_1} \{b_{1j}(t)\} > r \frac{a_2}{a_1} \max_{j=1, \dots, N_1} \{b_{2j}(t)\}\right]}{E\left[\max_{j=1, \dots, N_2} \{b_{2j}(t)\} \mid \max_{j=1, \dots, N_2} \{b_{2j}(t)\} > r^{-1} \frac{a_1}{a_2} \max_{j=1, \dots, N_1} \{b_{1j}(t)\}\right]}$$

or with

$$r' = r \frac{a_2}{a_1},$$

the fixed point equation becomes

$$r' = \frac{P\left\{\max_{j=1, \dots, N_1} \{b_{1j}(t)\} > r' \max_{j=1, \dots, N_1} \{b_{2j}(t)\}\right\}}{P\left\{\max_{j=1, \dots, N_2} \{b_{2j}(t)\} > (r')^{-1} \max_{j=1, \dots, N_1} \{b_{1j}(t)\}\right\}} \times \frac{E\left[\max_{j=1, \dots, N_1} \{b_{1j}(t)\} \mid \max_{j=1, \dots, N_1} \{b_{1j}(t)\} > r' \max_{j=1, \dots, N_1} \{b_{2j}(t)\}\right] N_2}{E\left[\max_{j=1, \dots, N_2} \{b_{2j}(t)\} \mid \max_{j=1, \dots, N_2} \{b_{2j}(t)\} > (r')^{-1} \max_{j=1, \dots, N_1} \{b_{1j}(t)\}\right] N_1}$$

which is independent of (a_1, a_2) . Hence, it is seen that first the result for variability can be obtained with arbitrary (a_1, a_2) , and then the distance effect can be obtained from

$$r = r' \frac{a_1}{a_2}$$

Note that the second numerical example ($a_1 = 1, a_2 = 10$) displays this separability.

5. Concluding Remark

The results use a number of simplifying assumptions (including no prediction errors and rate quantization, and always full buffers) and should be regarded as first approximations to provide insight and to help interpret more accurate simulations. Comparison with some simulations show that, although there are numerical differences depending on specific parameters, the analysis well characterizes the results qualitatively.

ACKNOWLEDGEMENT

The comments of Boris Tsybakov are appreciated.

REFERENCES

- [1] A. Goldsmith and P. Varaiya, "Increasing spectral efficiency through power control," ICC'93.
- [2] R. Knopp and P. Humblet, "Multiple-accessing over frequency-selective fading channels," PIMRC'95.
- [3] S.W. Kim and A. Goldsmith, "Truncated power control in code division multiple access communications," Globecom'97.
- [4] D. Tse, "A proportional fair forward-link scheduling algorithm," to appear.
- [5] J.M. Holtzman, "CDMA forward link waterfilling power control," Proc. VTC2000-Spring, pp. 1663-1667, May, 2000.
- [6] N. Joshi, S.R. Kadaba, S. Patel, G.S. Sundaram, "Downlink scheduling in CDMA data networks," ACM Mobicom2000.
- [7] H.A. David, "Order Statistics," 2d Ed., Wiley, 1981.
- [8] G.L. Stuber, "Principles of Mobile Communication," Kluwer, 1996.

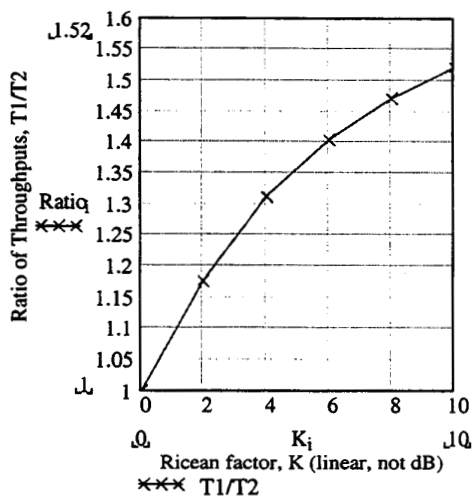


Figure 1—Throughput Ratio vs. Ricean Factor K, $a_1=a_2=1$, $N_1=N_2=5$

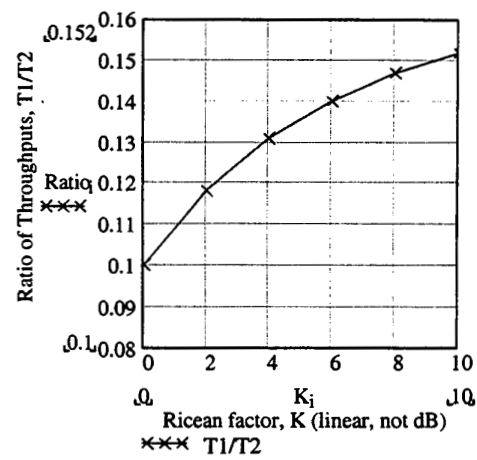


Figure 3—Throughput Ratio vs. Ricean Factor K, $a_1=1$, $a_2=10$, $N_1=N_2=5$

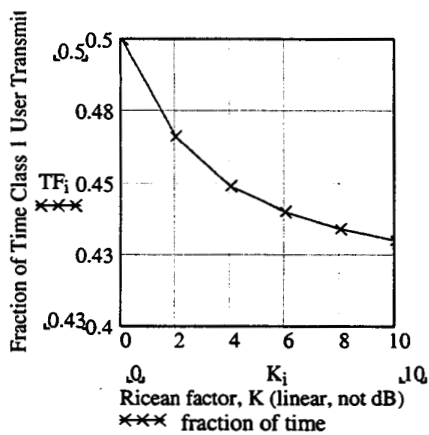


Figure 2 – Fraction of Time that User of Class 1 Transmits, $a_1=a_2$, $N_1=N_2=5$

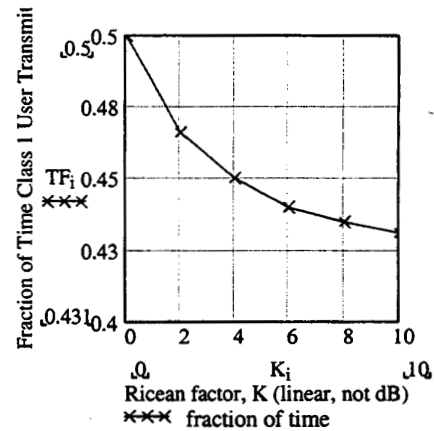


Figure 4 – Fraction of Time that User of Class 1 Transmits, $a_1=1$, $a_2=10$, $N_1=N_2=5$